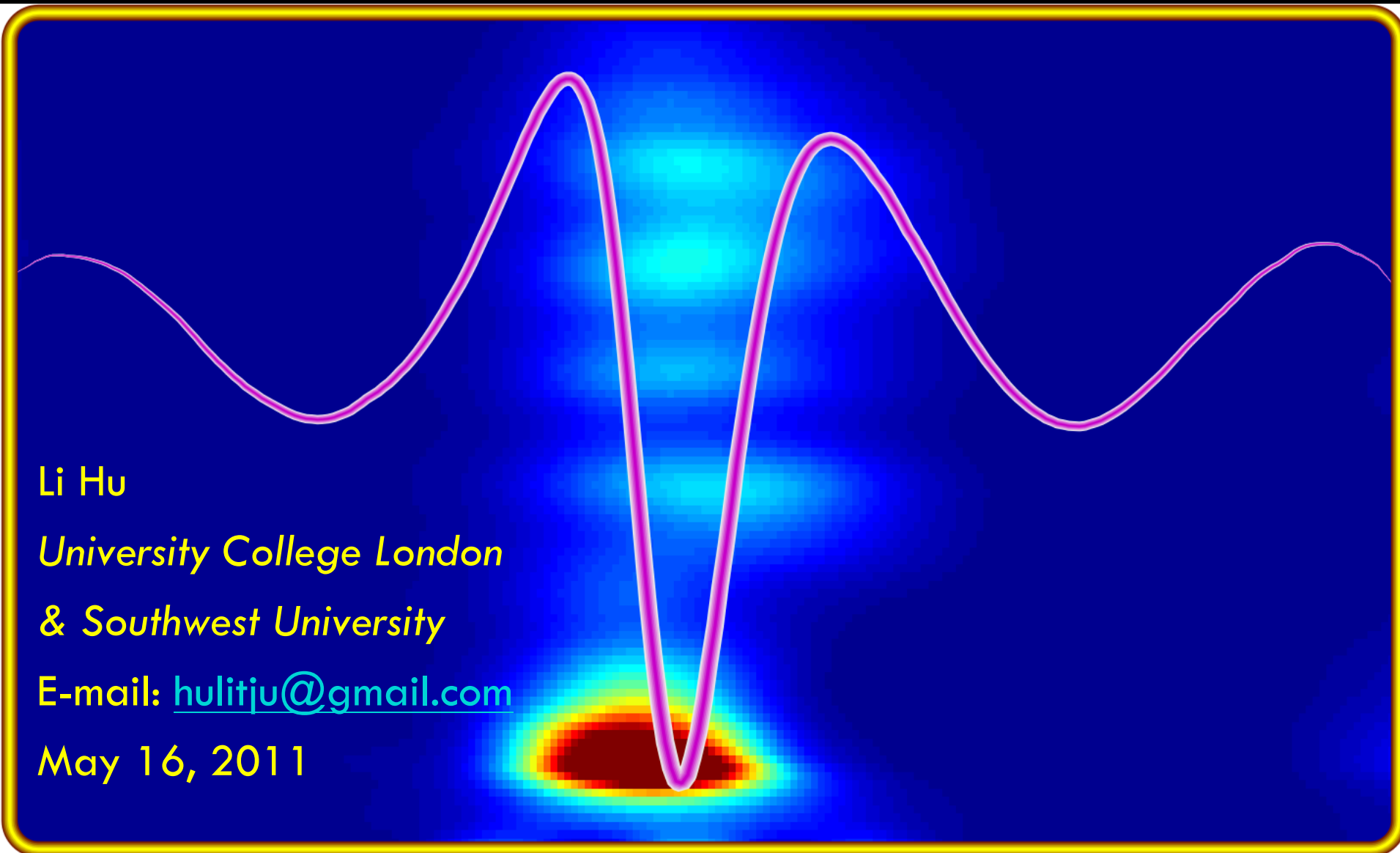


SINGLE TRIAL DETECTION TOOLBOX FOR EPILEPSY/EVOKED POTENTIALS (VERSION 1.0)—STEP 1



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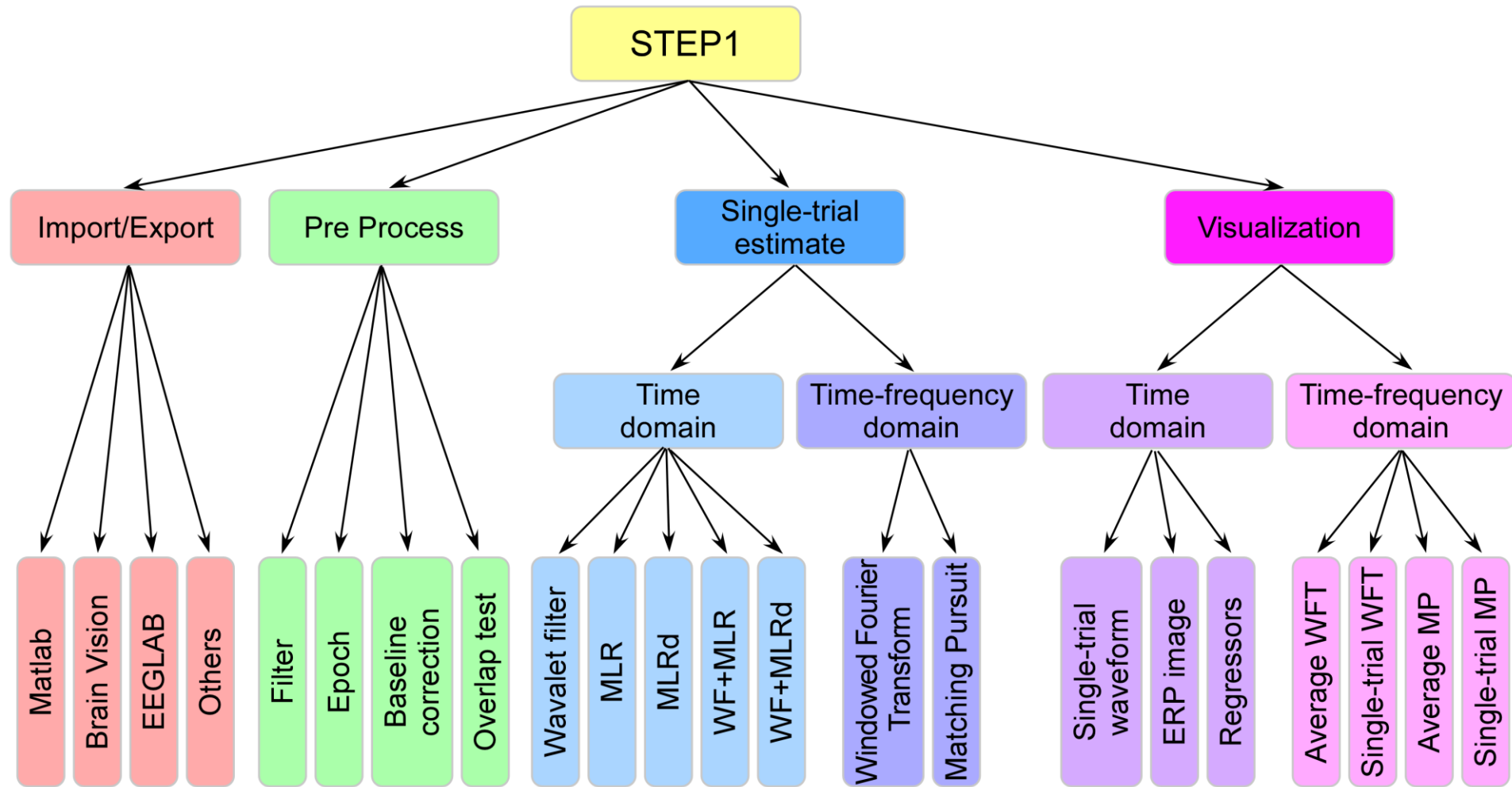
E-mail: hulitju@gmail.com

May 16, 2011

STEP 1—Introduction

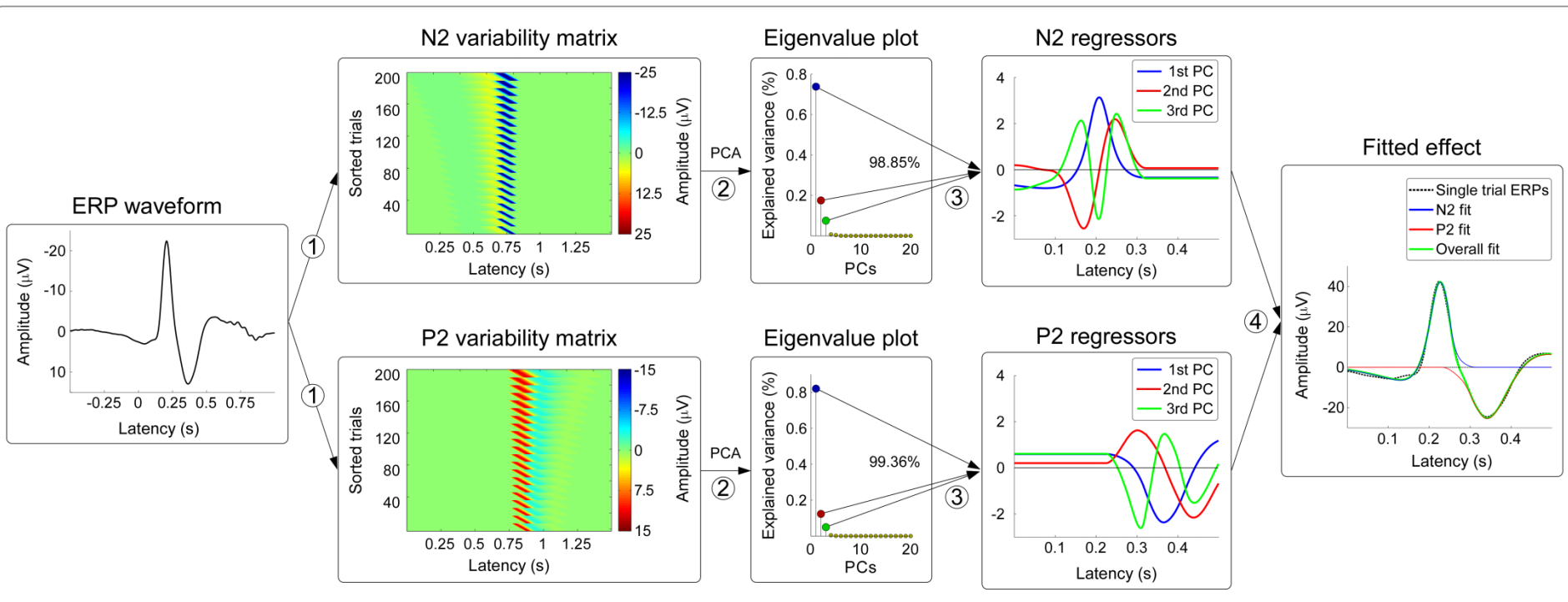
- Interactive Matlab toolbox to estimate single-trial EEG (ERP/Epilepsy) features
 - ▣ in the time domain
 - ▣ in the time-frequency domain
- Fully compatible with EEGLAB
- The estimated single-trial features
 - ▣ to explore the dynamics of spikes
 - ▣ To jointly with fMRI

STEP1—Functions

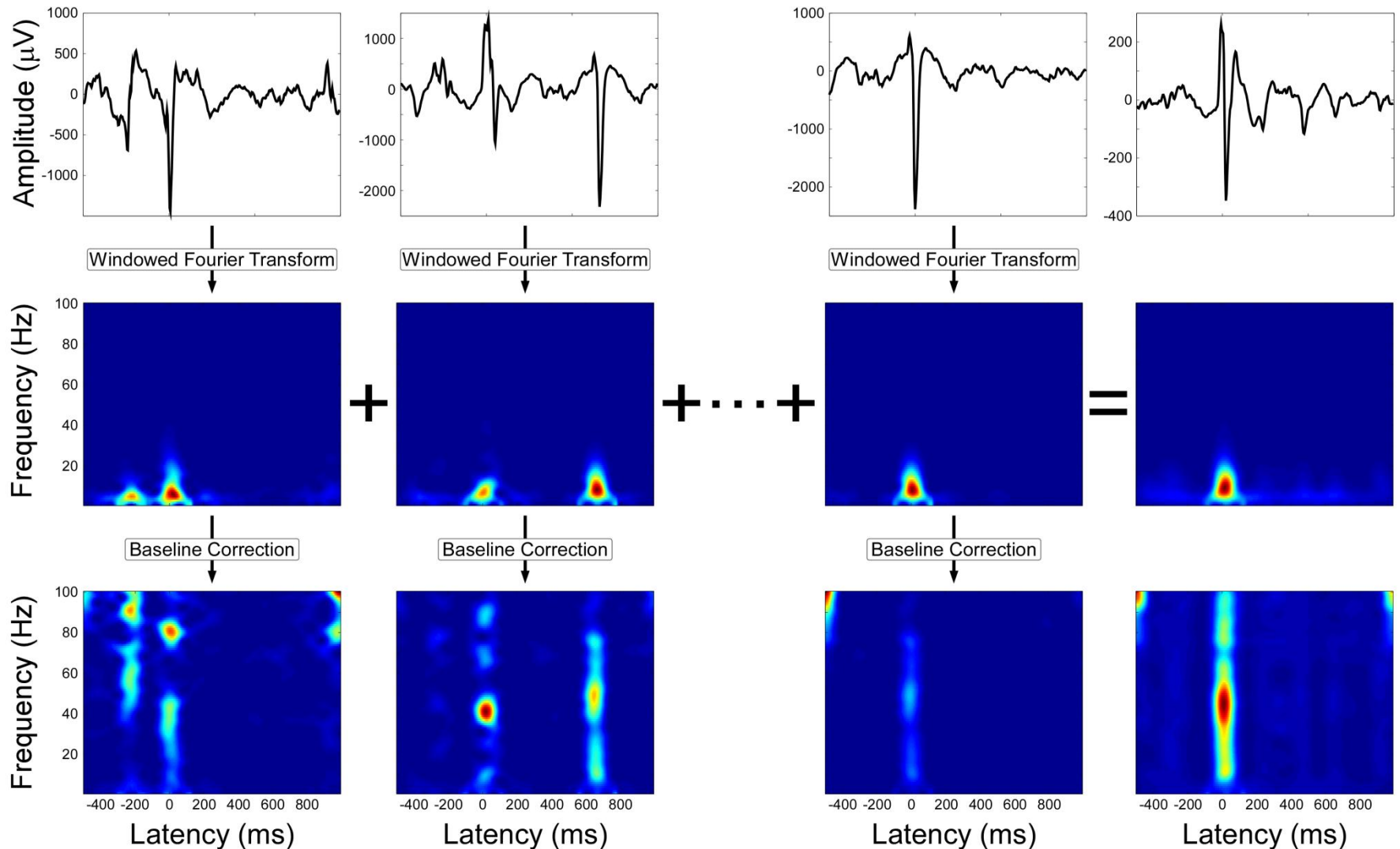


STEP1—MLRd principle

Multiple linear regression with dispersion term (MLR_d)

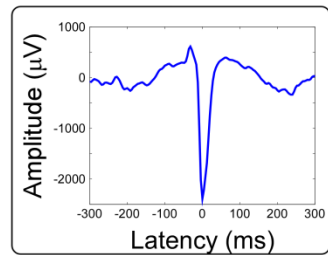


STEP1—WFT principle



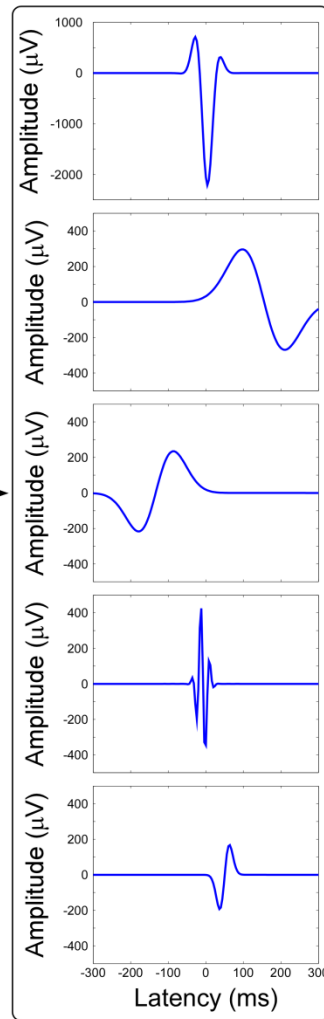
STEP1—MP principle

Single-trial waveform



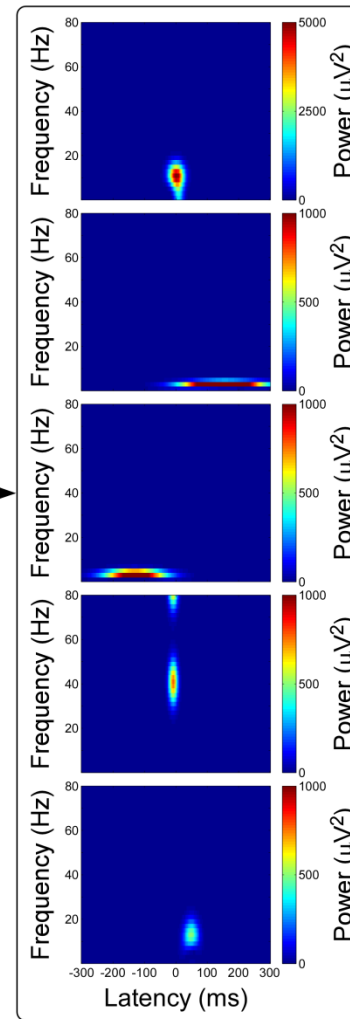
Matching Pursuit

Detailed waveforms



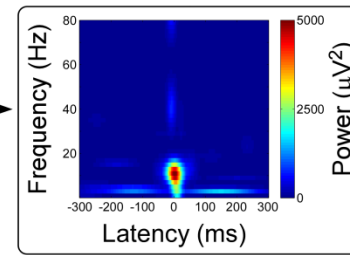
Time-frequency
Transform

Detailed TFDs



Average

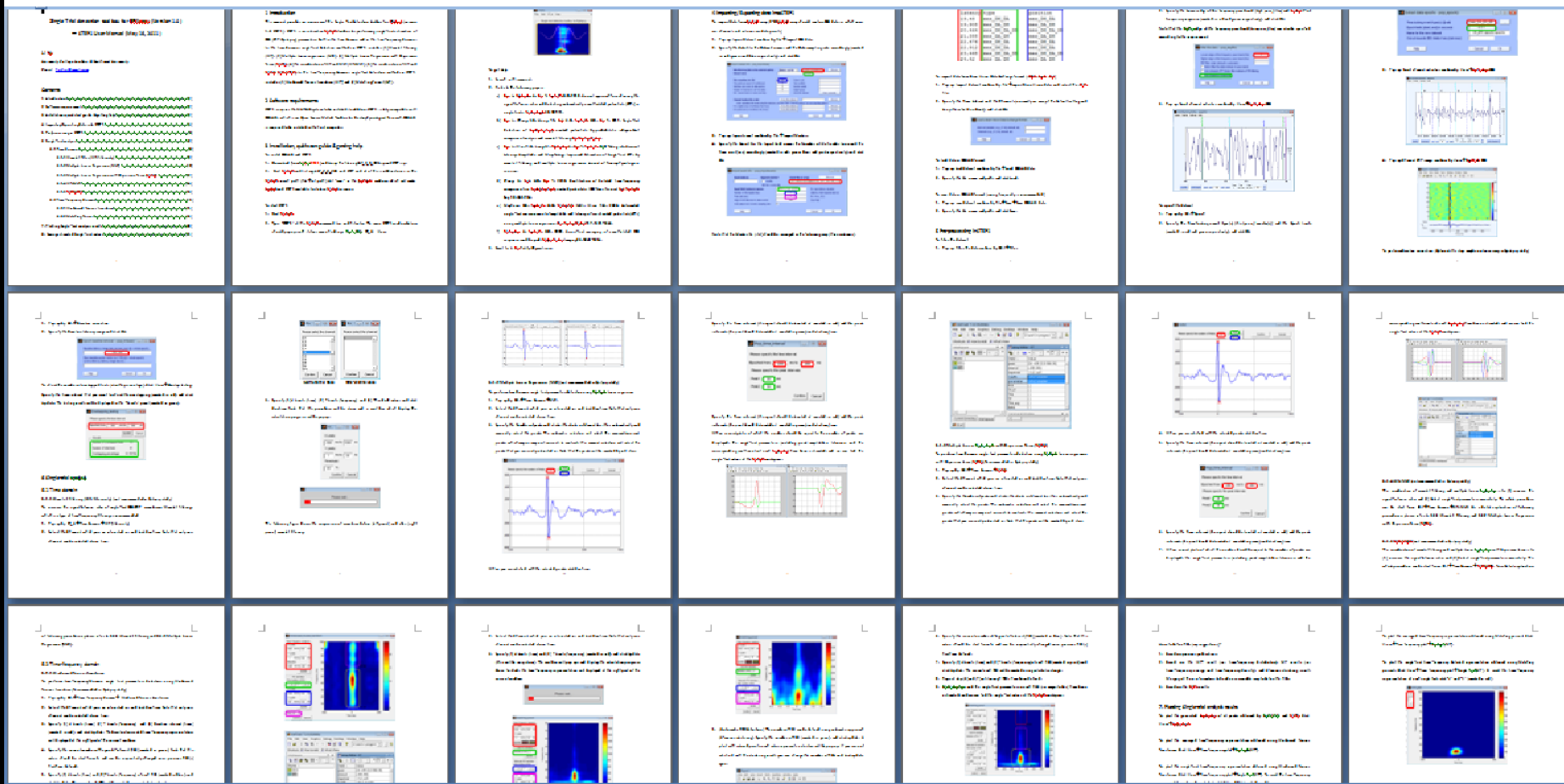
Overall TFD



STEP1—Manual

Single Trial detection toolbox for EPILEpsy (Version 1.0)

— STEP1 User Manual (May 16, 2011)



STEP 1—Processing procedure summary

- Install and start (Matlab, EEGLAB, STEP1)
- Import EEG data (data and marker files)
- Pre-processing (filter, epoch, overlap test)
- Single-trial analysis
 - Time domain MLRd
 - Time-frequency domain WFT
 - Time-frequency domain MP
- View and Save results (should be frequently)

STEP1—How to get help?

- Search in the manual
- Email to Li Hu: hulitju@gmail.com
- Refer to the following papers:
 - Hu, L., Mouraux, A., Hu, Y., Iannetti, G.D., 2010. A novel approach for enhancing the signal-to-noise ratio and detecting automatically event-related potentials (ERPs) in single trials. *Neuroimage* 50, 99-111.
 - Hu, L., Zhang, Z.G., Hung, Y.S., Luk, K.D., Iannetti, G.D., Hu, Y., 2011. Single-trial detection of somatosensory evoked potentials by probabilistic independent component analysis and wavelet filtering. *Clin Neurophysiol*.
 - Hu, L., Wise, R.G., Liang, M., Mouraux, A., Hu, Y., Iannetti, G.D. Taking into Account Latency, Amplitude and Morphology: Improved Estimation of Single-trial ERPs by wavelet filtering and multiple linear regression, *Journal of Neurophysiology*, in revision.
 - Zhang, Z., Luk, K.D., Hu, Y., 2010. Identification of detailed time-frequency components in somatosensory evoked potentials. *IEEE Trans Neural Syst Rehabil Eng* 18, 245-254.
 - Mayhew, S.D., Iannetti, G.D., Woolrich, M.W., Wise, R.G., 2006. Automated single-trial measurement of amplitude and latency of laser-evoked potentials (LEPs) using multiple linear regression. *Clin Neurophysiol* 117, 1331-1344.
 - Mouraux, A., Iannetti, G.D., 2008. Across-trial averaging of event-related EEG responses and beyond. *Magn Reson Imaging* 26, 1041-1054.

STEP 1—Thanks and case practice

- Thanks for your time/comments/suggestions!
- Functions will be advanced & updated in the future
- If there is any bug, please send it to my email address: hulitju@gmail.com
- Case practice!